

Christian_Herrera_HW6

March 2, 2026

1 Homework 6: OOP-2 (Solutions)

2 Assignment Submission Guidelines

Please follow the guidelines below for submitting your assignment:

1. Submission Deadline:

- All assignments must be submitted **no later than 23:59 PM next Tuesday (03/03)**.
- Late submissions will not be accepted unless prior arrangements have been made by the TAs.

2. Complete Your Work:

- Finish your work in Google Colab and ensure that your notebook is saved.

3. Submit Your Assignment on Canvas:

- Log in to **Canvas** and navigate to the assignment submission page.

4. File Naming Convention:

- Please name your files as follows: `Lastname_Firstname_AssignmentName`
- Example: `Alex_John_HW6.ipynb` and `Alex_John_HW6.pdf`

5. Technical Issues:

- If you encounter any technical issues with Canvas or your submission, please contact the TAs immediately **before the deadline** to avoid penalties.

2.1 Space Adventure (100 Points)

2.1.1 Objective:

You have to extend the space mission by adding different types of spaceships and astronauts. The goal is to use **inheritance** to create specialized subclasses for a more advanced space mission. You'll implement classes with attributes and methods, then override them in subclasses to customize the behavior of each type of spaceship and astronaut.

Tasks:

1. Spaceship Class (Superclass) (20 Points):

- Create a superclass named **Spaceship** with the following attributes:
 - **name** (string): The name of the spaceship.
 - **fuel** (integer): The spaceship's starting fuel, initialized to 100.
- Implement the following methods:

- `display_status()`: Prints the spaceship's name and fuel level.
 - `travel(distance)`: Reduces fuel based on the `distance` traveled. For every 10 units of distance, reduce fuel by 5 units. If fuel reaches zero, print that the spaceship has run out of fuel.
 - `refuel()`: Refuels the spaceship by 20 units, but the fuel cannot exceed 100.
2. **CargoSpaceship Class (Subclass of Spaceship) (20 Points):**
- Create a subclass named `CargoSpaceship` that inherits from `Spaceship` with an additional attribute:
 - `cargo_capacity` (integer): The maximum amount of cargo the spaceship can carry.
 - Override the `display_status()` method to also print the cargo capacity.
 - Add a new method `load_cargo(amount)` to load cargo. If the cargo exceeds the capacity, print a warning message.
3. **Astronaut Class (Superclass) (20 Points):**
- Create a superclass named `Astronaut` with the following attributes:
 - `name` (string): The name of the astronaut.
 - `oxygen_level` (integer): The astronaut's oxygen level, initialized to 100.
 - Implement the following methods:
 - `display_info()`: Prints the astronaut's name and oxygen level.
 - `perform_spacewalk()`: Decreases oxygen level by 20. If oxygen falls below 20, print a warning that the astronaut is running out of oxygen.
 - `replenish_oxygen()`: Increases oxygen by 20 units, but it cannot exceed 100.
4. **EngineerAstronaut Class (Subclass of Astronaut) (20 Points):**
- Create a subclass named `EngineerAstronaut` that inherits from `Astronaut` with an additional attribute:
 - `repair_skill` (integer): The astronaut's ability to repair the spaceship, initialized to 50.
 - Override the `perform_spacewalk()` method to also print the astronaut's repair skill after the spacewalk.
 - Add a new method `repair(spaceship)` to repair the spaceship's fuel by 10 units, but fuel cannot exceed 100.
5. **Mission Simulation (20 Points):**
- Implement a `space_mission()` function that:
 - Creates one `CargoSpaceship` and one `EngineerAstronaut` object.
 - The spaceship must travel a total distance of 50 units.
 - At the midpoint (25 units), the spaceship will need to refuel, and the astronaut will perform a spacewalk and repair the spaceship.
 - The mission ends when the spaceship completes the 50-unit journey or if the spaceship runs out of fuel or the astronaut runs out of oxygen.

2.1.2 Example Output:

```
[ ]: Spaceship 'Cargo Carrier' is ready for the mission!
Cargo Capacity: 500

Traveling to the distant planet...

Traveling 25 units...
```

Fuel remaining: 87

Astronaut Alice performs a spacewalk. Oxygen level: 80, Repair skill: 50

Refueling the spaceship...

Fuel is now 100

Repairing the spaceship... Fuel increased by 10 units. Fuel is now 100

Traveling another 25 units...

Fuel remaining: 87

Mission Complete! You have reached the planet!

2.1.3 Solution:

```
[10]: class Spaceship:
    def __init__(self, name, fuel=100):
        self.name = name
        self.fuel = fuel

    def display_status(self):
        print(f'Spaceship name: {self.name}')
        print(f'Fuel Level: {self.fuel}')

    def travel(self, distance):
        self.fuel -= distance * 0.5
        if self.fuel <= 0:
            self.fuel = 0
            print('You ran out of gas. ')

    def refuel(self):
        self.fuel += 20
        if self.fuel > 100:
            self.fuel = 100

class CargoSpaceship(Spaceship):
    def __init__(self, name, cargo_capacity, fuel=100):
        super().__init__(name, fuel)
        self.cargo_capacity = cargo_capacity
        self.cargo_loaded = 0

    def display_status(self):
        print(f'Spaceship name: {self.name}')
        print(f'Fuel Level: {self.fuel}')
        print(f'Cargo Capacity: {self.cargo_capacity}')
```

```

        print(f'Cargo Loaded: {self.cargo_loaded}')

    def load_cargo(self, amount):
        self.cargo_loaded += amount
        if self.cargo_loaded > self.cargo_capacity:
            print('Warning, you are overweight. ')

class Astronaut:
    def __init__(self, name, oxygen_level=100):
        self.name = name
        self.oxygen_level = oxygen_level

    def display_info(self):
        print(f'Astronaut name: {self.name}')
        print(f'Oxygen Level: {self.oxygen_level}')

    def perform_spacewalk(self):
        self.oxygen_level -= 20
        if self.oxygen_level < 0:
            self.oxygen_level = 0
        if self.oxygen_level < 20:
            print("The astronaut\'s oxygen level is low. ")

    def replenish_oxygen(self):
        self.oxygen_level += 20
        if self.oxygen_level > 100:
            self.oxygen_level = 100

class EngineerAstronaut(Astronaut):
    def __init__(self, name, oxygen_level=100, repair_skill=50):
        super().__init__(name, oxygen_level)
        self.repair_skill = repair_skill

    def perform_spacewalk(self):
        self.oxygen_level -= 20
        if self.oxygen_level < 0:
            self.oxygen_level = 0
        if self.oxygen_level < 20:
            print("The engineer astronaut\'s oxygen level is low. ")
        print(f'Repair Skill: {self.repair_skill}')

    def repair(self, spaceship):
        spaceship.fuel += 10
        if spaceship.fuel > 100:
            spaceship.fuel = 100

```

```

        print('Engineer has repaired the spaceship. +10 fuel.')

def space_mission():
    ship = CargoSpaceship('Atlas', cargo_capacity=100)
    astro = EngineerAstronaut('Christian')
    total_distance = 50
    traveled = 0
    print('Start mission!')
    ship.display_status()
    astro.display_info()

    while traveled < total_distance:
        step = 25 if traveled == 0 else 25
        print(f'Traveling {step} units. ')
        ship.travel(step)
        traveled += step
        ship.display_status()

        if traveled == 25:
            print('Midpoint section')
            print('Refueling spaceship.')
            ship.refuel()
            print('Astronaut performing spacewalk.')
            astro.perform_spacewalk()
            print('Astronaut repairing spaceship.')
            astro.repair(ship)
            ship.display_status()
            astro.display_info()

    print('Mission Complete.')

space_mission()

```

```

Start mission!
Spaceship name: Atlas
Fuel Level: 100
Cargo Capacity: 100
Cargo Loaded: 0
Astronaut name: Christian
Oxygen Level: 100
Traveling 25 units.
Spaceship name: Atlas
Fuel Level: 87.5
Cargo Capacity: 100
Cargo Loaded: 0
Midpoint section

```

Refueling spaceship.
Astronaut performing spacewalk.
Repair Skill: 50
Astronaut repairing spaceship.
Engineer has repaired the spaceship. +10 fuel.
Spaceship name: Atlas
Fuel Level: 100
Cargo Capacity: 100
Cargo Loaded: 0
Astronaut name: Christian
Oxygen Level: 80
Traveling 25 units.
Spaceship name: Atlas
Fuel Level: 87.5
Cargo Capacity: 100
Cargo Loaded: 0
Mission Complete.

[]: